

Soumyajit Bhandary

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PROFESSIONAL SUMMARY

- Information Technology graduate with hands-on experience in backend engineering, distributed systems, Generative AI, Agentic AI and LLMs. Proficient in building production-grade scalable applications, RAG pipelines. Seeking roles in GenAI/LLM Application or AI-Backend Engineering.

EDUCATION

- Kalinga Institute of Industrial Technology** Bhubaneswar, India
Bachelor of Technology - Information Technology; GPA: 8.01
Courses: Operating Systems, Data Structures, Object Oriented Programming, Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Networking, Databases
Sep 2022 - May 2026

PROJECTS

ReconcileIQ — Agentic AI Invoice Reconciliation System

[GitHub](#)

Python, FastAPI, PostgreSQL, pgvector, Redis, OpenAI API, Langfuse, Docker

- Designed and built a multi-agent pipeline (router, reconciliation planner, policy gate, async vendor communication agent) that automates 3-way invoice matching, cutting manual AP reconciliation from a multi-day process to sub-2-second automated resolution for the majority of cases.
- Engineered a durable, queue-backed async workflow (Redis + PostgreSQL) with idempotent transaction handling, verified zero data loss under process failure and restart.
- Implemented LLM observability and automated evals (Langfuse tracing, golden-dataset scoring) to detect, diagnose, and validate prompt regressions, enabling continuous model improvement.
- Enforced human-in-the-loop controls via a policy gate (dollar-threshold-based escalation) and mandatory approval step before any financial posting, with idempotency keys preventing duplicate transactions on retry.
- Built a full-stack review dashboard (Flask + vanilla JS) for human case triage and approval, and load-tested the system end-to-end (p50: 1.56s, p95: 1.95s across the LLM-bound path).

AutoNexus AI — Multi-Agent LLM Analytics Platform

[GitHub](#)

LangGraph, FastAPI, PostgreSQL, React

- Architected a multi-agent orchestration system using LangGraph StateGraph, coordinating 3 parallel LLM agents with conditional fan-out/fan-in routing and Pydantic-enforced structured outputs, reducing unnecessary agent invocations by routing only relevant analyses per request.
- Implemented a human-in-the-loop approval workflow using LangGraph's interrupt()/Command(resume) primitives with PostgreSQL checkpointing, verified via automated tests to persist and resume correctly across full process restarts.
- Built a FastAPI backend and React frontend with a sandboxed, read-only SQL data-access layer and a deterministic validation pipeline enforcing citation and confidence checks on all LLM-generated outputs.
- Validated system reliability through a 30-test automated suite (pytest) and load testing (Locust) at 1,285 requests with 0% failure rate (p95: 1.7s, p99: 5.4s).

RailBook — Production-Grade Train Booking System

[GitHub](#)

Python, FastAPI, PostgreSQL, Redis, Elasticsearch, RabbitMQ, Celery, Next.js

- Architected a scalable, production-grade IRCTC-style booking platform following distributed system design principles - addressing real concurrency and correctness problems: double-booking races, stale cache, and instant JWT revocation.
- Built read-through Redis caching with surgical key invalidation on writes (keyed by route+date+class); decoupled email delivery from the HTTP response via Celery + RabbitMQ — POST /bookings returns 201 in 50ms with confirmation sent async with exponential back-off. Overall system design was built to support horizontal scaling with stateless API layer and Redis as shared cache.
- Validated under load — 200 concurrent booking requests with zero duplicate PNRs via Locust stress test.

SKILLS SUMMARY

- Generative AI & LLMs** : LLM Agents & Multi-Agent Orchestration, LangGraph, OpenAI API, Prompt Engineering, RAG/Vector Search, LLM Evaluation & Observability, Human-in-the-Loop (HITL) Workflow Design
- Languages** : Python , SQL, Java
- Data & Tools** : Redis, Elasticsearch, RabbitMQ, Celery, Docker, LangSmith, Tavily Search, BeautifulSoup
- Backend & Systems**: FastAPI, Flask, async/queue-backed workflows, idempotent transaction design, distributed systems (caching, cache invalidation, concurrency control), REST API design
- Database (Relational & Vector DB)** : PostgreSQL, MySQL, pgvector, FAISS, Chroma DB